

Overview of Artificial neural network

Subject: Artificial neural network

$$a^{(l)} = \sigma(W^{(l)}a^{(l-1)} + b^{(l)})$$

1.

A Brief Introduction to Neural Networks (D. Kriesel) – Illustrated, bilingual manuscript about neural networks; Topics so far: Perceptrons, Backpropagation, Radial Basis Functions, Recurrent Neural Networks, Self Organizing Maps, Hopfield Networks. Review of Neural Networks in Materials Science Archived 7 June 2015 at the Wayback Machine Artificial Neural Networks Tutorial in three languages (Univ. Politécnica de Madrid) Another introduction to NN Next Generation of Neural Networks Archived 24 January 2011 at the Wayback Machine – Google Tech Talks Performance of Neural Networks Neural Networks and Information Archived 9 July 2009 at the Wayback Machine Sanderson G (5 October 2017). "But what is a Neural Network?". 3Blue1Brown. Archived from the original on 7 November 2021 – via YouTube. Hakim MA, Alam MI (2025). "Biologically inspired neural network layer with homeostatic regulation and adaptive repair mechanisms". Scientific Reports. 15 (1). Nature Publishing Group: 33903. Bibcode:2025NatSR..1533903H. doi:10.1038/s41598-025-09114-8. ISSN 2045-2322. PMC 12484884. PMID 41028030.

2.

Natural language processing In natural language processing, NNs are used for tasks such as text classification, sentiment analysis, machine translation, to answer free-form questions, act as chatbots, and to summarize and analyze texts. This has implications for automated customer service, content moderation, and language understanding technologies.

3.

Self-learning Self-learning was introduced in 1982 along with a crossbar adaptive array (CAA) NN that could teach itself. It is a system with only one input, situation s , and only one output, action (or behavior) a . It has neither external advice input nor external reinforcement from the environment. The CAA computes, in a crossbar fashion, both decisions about actions and emotions (feelings) about encountered situations. The system is driven by the interaction between cognition and emotion. Given the memory matrix, $W = ||w(a,s)||$, the crossbar self-learning algorithm in each iteration performs the following computation:

4.

Concept drift The statistical properties of input data may change over time, a phenomenon known as concept drift or non-stationarity. Drift can reduce predictive accuracy and lead to unreliable or biased decisions if it is not detected and corrected. In practice, this means that the model's accuracy in deployment may differ substantially from levels observed during training. Several strategies have been developed to monitor neural networks for drift and degradation:

5.

Hardware Neural networks require enormous computing resources for training. While the brain requires only 20 watts of power, training commercial transformers requires (2026) data centers with hundreds of megawatts. From 1991 to 2015, computing power, especially as delivered by GPGPUs (on GPUs), increased around a million-fold, enabling the use of the backpropagation algorithm. Neuromorphic engineering or a physical neural network constructed non-von-Neumann chips to directly implement neural networks in circuitry. For example, Alphabet introduced custom chips (Tensor Processing Unit).

6.

Network NNs connect neurons to each other, using the output of some neurons as the input of others, akin to biological axon-synapse-dendrite connections. The network forms a directed, weighted graph. The weights apply to both the graph's the nodes and edges.

7.

Error-based monitoring: comparing current predictions against ground-truth labels. This approach directly quantifies predictive performance but may be impractical when labels are delayed or impractical to obtain. Data distribution monitoring: detecting changes in the input data distribution using statistical tests, divergence measures, or density-ratio estimation. Representation monitoring: tracking the distribution of internal embeddings or hidden-layer features. Shifts in the latent representation can indicate nonstationarity even when labels are unavailable. Statistical methods such as statistical process control charts have been adapted for this purpose.

8.

Computational power The multilayer perceptron is a universal function approximator, as proven by the universal approximation theorem. However, the proof does not specify the number of neurons required, the network topology, the weights or the learning parameters. A recurrent architecture with rational-valued weights (as opposed to full precision real number-valued weights) has the power of a universal Turing machine, using a finite number of neurons and linear connections. Further, the use of irrational values for weights results in a machine with super-Turing power.

9.

Dataset bias Neural networks are dependent on the quality of their training data; low quality data can lead the model to produce poor results. Biased data produces biased results, requiring trainers to detect and correct them. For example, data that underrepresents some demographic groups may prevent cause errors, e.g., in facial recognition and law enforcement. In 2018, Amazon scrapped a recruiting tool because the model favored men over women for jobs in software engineering due to the larger number of male workers in the field. The system penalized resumes with the word "woman" or the name of a women's college. One corrective is to add synthetic data to offset the bias.

10.

Medicine NNs analyze medical datasets. They enhance diagnostic accuracy, especially by interpreting complex medical imaging for early disease detection, and by predicting patient outcomes for personalized treatment planning. In drug discovery, NNs speed up the identification of potential drug

candidates and predict their efficacy and safety, significantly reducing development time and costs. Additionally, their application in personalized medicine and healthcare data analysis allows tailored therapies and efficient patient care management.

11.

Applications that must generalize well to unseen examples, must avoid over-training. This arises in convoluted or over-specified systems when the network capacity is much larger than needed. Two approaches address over-training. Cross-validation and similar techniques can check for over-training and select appropriate hyperparameters to minimize generalization error. Regularization in a probabilistic (Bayesian) framework can be performed by selecting a larger prior probability over simpler models; but also in statistical learning theory, where the goal is to minimize two quantities: the 'empirical risk' and the 'structural risk', which roughly corresponds to the error over the training set and the predicted error in unseen data due to overfitting.

12.

Modern deep learning Between 2009 and 2012, NNs began winning prizes in image recognition contests, approaching human performance on various tasks, initially in pattern recognition and handwriting recognition. In 2011, DanNet, a CNN by Dan Ciresan, Ueli Meier, Jonathan Masci, Luca Maria Gambardella, and Schmidhuber achieved for the first time superhuman performance in a visual pattern recognition contest, outperforming traditional methods by a factor of 3. It then won more contests. They also showed how max-pooling CNNs on GPU improved performance. In October 2012, AlexNet by Alex Krizhevsky, Ilya Sutskever, and Hinton won the large-scale ImageNet competition by a significant margin over shallow machine learning methods. Further incremental improvements included the VGG-16 network by Karen Simonyan and Andrew Zisserman and Google's Inceptionv3. In 2012, Ng and Dean created a network that learned to recognize higher-level concepts, such as cats, from training only on unlabeled images. Unsupervised pre-training and increased computing power from GPUs and distributed computing allowed the use of larger networks, particularly in image and visual recognition problems, which became known as "deep learning". Radial basis function and wavelet networks were introduced in 2013. These can be shown to offer best approximation properties and have been applied in nonlinear system identification and classification applications. Generative adversarial networks (GANs) (Ian Goodfellow et al., 2014) became state of the art in generative modeling from 2014–2018. GAN was originally published in 1991 by Schmidhuber, who called it "artificial curiosity": two neural networks contest with each other in the form of a zero-sum game, where one network's gain is the other network's loss. The first network is a generative model that models a probability distribution over output patterns. The second network learns by gradient descent to predict the reactions of the environment to these patterns. Excellent image quality was achieved by Nvidia's StyleGAN (2018) based on the Progressive GAN by Tero Karras et al. Here, the GAN generator is grown from small to large scale in a pyramidal fashion. Image generation by GAN reached popular success, and provoked discussions concerning deepfakes. Diffusion models (2015) eclipsed GANs in generative modeling thereafter, with systems such as DALL·E 2 (2022) and Stable Diffusion (2022). In 2014, the state of the art was training "[a] very deep neural network" with 20 to 30 layers. Stacking too many layers led to a steep reduction in training accuracy, known as the "degradation" problem. In 2015, training very deep networks advanced with the highway network, published in May, and the residual neural network (ResNet) in December. ResNet behaves like an open-gated Highway Net.